Predict Spatial-Resolution Grid for One Age

Description

Builds abiotic and optional spatial predictors for one age slice and returns long-format taxon occurrence probabilities.

Usage

```
predict_spatial_resolution_grid_age(
  prediction_inputs,
  data_grid,
  data_grid_coords_projected,
  age,
  abiotic_variables,
  x_lim,
  y_lim,
  cache_dir,
  spatial_mode = "spatiotemporal",
  climate_fn = NULL,
  spatial_interpolate_fn = NULL,
  spatial_basis_project_fn = NULL,
  spatiotemporal_interpolate_fn = NULL,
  predict_fn = NULL
)
```

Arguments

`prediction_inputs`	Named list returned by [load_spatial_resolution_prediction_inputs()].
`data_grid`	Prediction grid with 'grid_id', 'coord_long', and 'coord_lat'.
`data_grid_coords_projected`	Projected prediction grid coordinates with row names 'grid_<id>'.

`age`	Numeric scalar age in years before present.
`abiotic_variables`	Character vector of CHELSA variables to extract.
`x_lim`	Numeric longitude range.
`y_lim`	Numeric latitude range.
`cache_dir`	Existing CHELSA cache directory.
`spatial_mode`	Spatial predictor mode: '"spatial"', '"spatiotemporal"', or '"none"'.
`climate_fn`	Climate extraction function. If 'NULL' (default), [extract_prediction_climate()] is used.
`spatial_interpolate_fn`	Spatial interpolation function. If 'NULL' (default), [interpolate_mev_to_grid()] is used.
`spatial_basis_project_fn`	Reusable-basis projection function. If 'NULL' (default), [project_spatial_mev_basis()] is used when basis state is available.
`spatiotemporal_interpolate_fn`	Spatiotemporal interpolation function. If 'NULL' (default), [interpolate_st_mev_to_grid()] is used.
`predict_fn`	Prediction function. If 'NULL' (default), the function resolves 'sjSDM:::predict.sjSDM'.

Value

Long tibble with predicted probabilities by grid cell, age, and taxon.

Examples

```
## Not run:

predict_spatial_resolution_grid_age(

  prediction_inputs = prediction_inputs,

  data_grid = data_grid,

  data_grid_coords_projected = data_grid_coords_projected,
```

```
  age = 0,
  abiotic_variables = c("bio1", "bio12"),
  x_lim = c(-10, 40),
  y_lim = c(35, 70),
  cache_dir = "Data/Temp/chelsa/prediction"
)

## End(Not run)
```